Random Variables

A random variable is a variable that takes on different numerical values, each associated with a probability. It is important to note that random variables can be discrete or continuous. In this lecture, we focus on discrete random variables.

Discrete Random Variables

↳ Binomial distribution

Bernoulli distribution

Examples of random variables:
The number of heads in a series of coin tosses.
The sum of the numbers rolled on two dice.

Probability

0.5 ← 1/2 —— H ⟶ 1 $

0.5 ← 1/2 —— T ⟶ -1 $

Continuous Random Variables

temp at your home ⟶

Normal distribution

standard Normal distribution

Random variables

- Discrete ↓ Binomial, Bernoulli
- Continuous ↓ Normal, Standard Normal

Probability distribution function (PDF)

- **Discrete** → PMF (Probability mass function)
- **Continuous** → PDF (Probability density function)

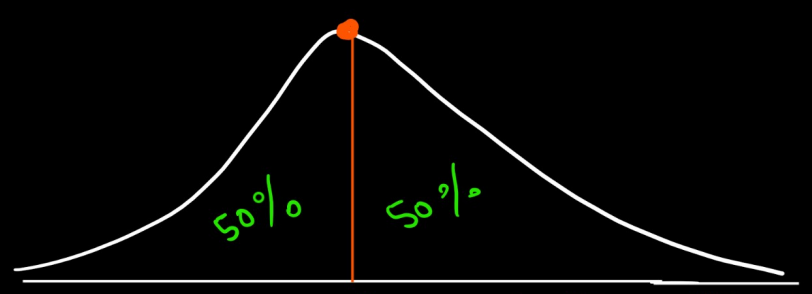
$$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} .$$

↑ Normal Distribution

$\sigma \longrightarrow$ standard deviation

$\mu \longrightarrow$ mean

Normal / Gaussian / Symmetric Distribution

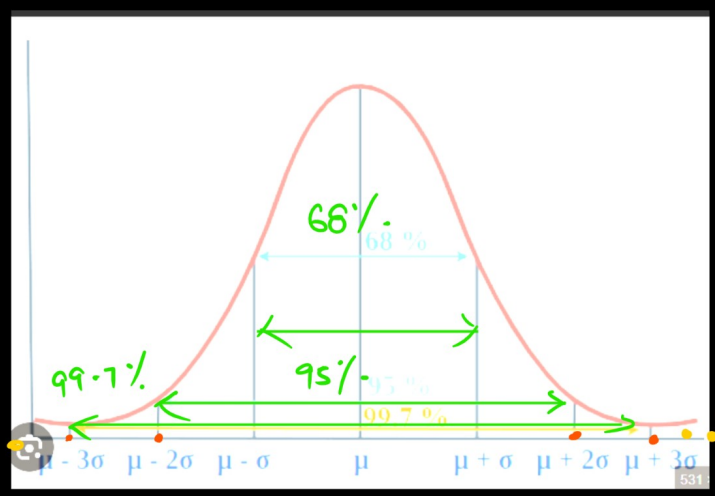


Mean = Median = Mode

SNF $\longrightarrow$ $\mu = 0, \ \sigma = 1$

standardization

Empirical Rule $\longrightarrow$ to detect the outliers



outlier

$-3 \quad -2 \quad \mu-\sigma \ (-1) \quad \mu \quad \mu+\sigma \ (+1) \quad +2 \quad +3$